

Domain-Specific Financial LLMs

Lecture 08 — When Finance Needs Its Own Models

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- 1 Why Domain-Specific LLMs?
- 2 A Taxonomy of Financial LLMs
- 3 Pre-training Strategies for Finance
- 4 Benchmark Comparisons
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Learning Objectives

By the end of this lecture you should be able to:

- 1 Explain *why* general-purpose LLMs underperform on financial text, and name the linguistic features that create the gap.
- 2 Place any financial model in a taxonomy: encoder-only vs. decoder-only, from-scratch vs. domain-adaptive vs. fine-tune-only.
- 3 Apply **domain-adaptive pre-training (DAPT)** and design a balanced corpus.
- 4 Read FinQA / FLUE / FinBen results in terms of the task taxonomy.
- 5 Weigh licensing, compute, governance, and accuracy when choosing a model for production.

THE BIGGER PICTURE

General models have *breadth* — hundreds of billions of web tokens give them world knowledge about options, CDS, and Basel III. Finance demands *specificity* — a model that confuses “basis risk” with “basis points”, misreads an income-statement table, or reads “liability” with its everyday connotation produces outputs that are not merely imprecise but **dangerous** downstream.

The resolution is not breadth *vs.* depth but *how to combine them*. Domain-specific financial LLMs orient pre-training, fine-tuning, or both toward the linguistic distribution of financial text.

First question: what actually makes financial language different?

Three structural features underrepresented in general corpora

- ❶ **Lexical specificity.** Common words carry precise, sometimes *reversed* meanings: “liability”, “short”, “haircut”. The Loughran–McDonald dictionary catalogues ~350 words whose polarity is neutral or inverted in SEC filings vs. general English.
- ❷ **Numeric density.** “Revenue grew 12.3% YoY to \$4.7B” is fluent English *and* an arithmetic claim. FinQA requires multi-step reasoning over tables from 10-Ks.
- ❸ **Tabular embedding.** Income statements, balance sheets, loan tapes encode meaning in 2-D grids; standard tokenisation flattens them and loses the row/column alignment.

A model is *financially literate* to the extent it resolves token meaning on all three dimensions.

Vocabulary Mismatch is a Tokenisation Problem

UNDER THE HOOD

The mismatch operates at the **sub-word** level. A BPE tokeniser trained on general text may split EBITDA into 4–5 sub-tokens, none of which carries financial meaning — the burden shifts onto the attention layers to reassemble it.

- Tokenisers trained on financial corpora represent abbreviations (EBITDA, CUSIP, LIBOR) as single tokens.
- BloombergGPT trained a *custom* BPE tokeniser on its combined general+financial corpus precisely so domain tokens stay compact.

Why it matters

Fewer tokens per financial term $\Rightarrow$ shorter sequences, lower compute, and crucially a single embedding the model can specialise — instead of a fragile composition over fragments.

How General Models Fail — Four Concrete Modes

Failure mode	Evidence
Sentiment polarity inversion	“strong”, “leverage”, “unprecedented” flip valence; general model fine-tuned on Financial PhraseBank trails FinBERT despite <i>more</i> parameters.
Numerical hallucination	GPT-class models fabricate EPS / figures that diverge from the source 10-K (Kang & Liu, 2023).
Table-parsing failure	On FinQA, GPT-3 (175B) zero-shot <60%; fine-tuned specialists >80%. Prompting alone does not close the gap.
Regulatory ambiguity	ISDA / Basel / SEC comment letters are a legal sub-dialect; SEC-BERT beats general BERT on NER and clause classification.

Common root

All four reflect the gap between the financial distribution and the web-text distribution that dominates pre-training. The remedy: shift the model’s prior toward finance.

Two Axes of Classification

THE BIGGER PICTURE

Every financial model is a point in a 2-D design space. Where you land dictates compute cost, capability profile, and deployment constraints.

Axis 1 — Architecture

- Encoder-only (bidirectional):
classification, extraction
- Decoder-only (autoregressive):
generation, multi-step reasoning
- Encoder-decoder (seq2seq):
summarisation, table-to-text

Axis 2 — Training strategy

- From scratch (BloombergGPT)
- Domain-adaptive continuation / DAPT (FinBERT)
- Fine-tune / instruction-tune only (FinMA, InvestLM)

Definition (Financial LLM)

A model is *financial* if its pre-training corpus has a significant financial fraction, *or* it is fine-tuned on financial NLP tasks, *or* its tokeniser is extended for financial vocabulary.

Encoder–Decoder: the T5 Family in Finance

UNDER THE HOOD

A bidirectional **encoder** reads the full source; an autoregressive **decoder** emits output token by token. The separate encoding pass lets the model attend to a long source *before* writing a single output token — reducing copy errors and improving factual consistency.

Canonical model: **T5** / instruction-tuned **FLAN-T5**. Financial uses:

- Earnings-call summarisation (hours of commentary → structured abstract)
- Abstractive 10-K summarisation (100 pages → one-page narrative)
- Financial table-to-text generation

When still preferred: constrained-output, extraction-heavy summarisation, even though large decoder-only models (FinMA) now also handle these at scale.

Encoder Family: FinBERT, SEC-BERT, FinBERT-tone

Model		Base / corpus	Specialisation
FinBERT	(Araci 2019)	BERT-base, ~1.8B words: Reuters TRC2, Bloomberg news, PhraseBank	3-class sentiment; large F1 gain over BERT-base on PhraseBank
FinBERT-tone	(Yang et al. 2020)	BERT-large, earnings calls + analyst reports + filings	LM tone: optimistic / pessimistic / litigious / uncertain / constraining
SEC-BERT	(Loukas 2022)	BERT-base, ~3B tokens of EDGAR only (10-K/Q, 8-K, proxies)	NER + clause classification on regulatory text

Why default to encoders for classification?

110–340M params, open weights, well-understood fine-tuning. EDGAR is a coherent legal sub-dialect — a news-trained model does *not* fully transfer to it.

Worked Example: FinBERT-tone on an Earnings Call

Transcript excerpt

“While we are **cautiously optimistic** about the trajectory of our core business, we want to flag meaningful **headwinds** in the European segment, driven by continued **softness** in consumer demand and heightened **competitive pressure**.”

General sentiment model

Labels **positive** — “optimistic” is a strong positive signal in product-review / social-media text.

FinBERT-tone

Labels **uncertain / pessimistic** — “cautiously optimistic” is a hedge; “headwinds”, “softness” are near-universally negative in this register.

Downstream stakes

This tone feeds analyst sentiment indices that drive systematic trading strategies (Tetlock 2007). A label error here is a position error.

Decoder Family: BloombergGPT, FinGPT, FinMA, InvestLM

Model	Scale	Approach
BloombergGPT	50B params, 708B tokens	From scratch; ~50/50 finance (363B FinPile) / general (345B Pile, Wiki, books). Beats peers on financial NLP, matches GPT-NeoX-20B on general tasks.
FinGPT	LoRA on LLaMA/ChatGLM/-Falcon	Open-source, <i>data-centric</i> ; fine-tune on a single A100 with live news APIs. Democratisation over peak accuracy.
FinMA (PIXIU)	7B LLaMA	Instruction-tuned on 136k financial instruction-response pairs across 5 task types. Strong in-distribution; weaker on unseen formats.
InvestLM	65B LLaMA	Instruction-tuned on a <i>small, curated</i> set (CFA Qs, research reports). Large base beats 7B models with fewer examples.

Decoder Family — the Autoregressive Objective

UNDER THE HOOD

Decoder LLMs are autoregressive: $\mathcal{L}_{\text{CLM}} = -\sum_t \log p_{\theta}(w_t \mid w_{<t})$. The roughly 50/50 corpus split in BloombergGPT is deliberate — a finance-only model loses the world knowledge that makes an LLM useful on finance tasks requiring general reasoning.

Reading the four models as a design spectrum

- **BloombergGPT** buys peak accuracy with from-scratch compute few can match.
- **FinGPT** trades peak accuracy for reproducibility on a single A100.
- **FinMA** is strong in-distribution but brittle on unseen formats.
- **InvestLM** shows a large base + small curated set can beat 7B models.

Decision rule

- **Fixed label space** (pos/neg/neutral, default/no-default)? $\Rightarrow$ start with an **encoder** (FinBERT, SEC-BERT): compact, fast, calibrated.
 - **Open-ended output** (draft a risk section, answer from a 10-K)? $\Rightarrow$ a **decoder** is necessary.
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- Encoders: 110–340M, fast to fine-tune, well-understood.
 - Decoders: an order of magnitude larger, more inference memory, harder to calibrate for structured prediction.
 - **Hybrid**: a small decoder + a well-indexed document store (RAG) can substitute for a much larger model (Lecture on LLM agents).

From Scratch or Continue? The Build Decision

THE BIGGER PICTURE

When Bloomberg built BloombergGPT it faced the choice every team confronts: train a new model on a mixed corpus, or start from a general model and *continue* pre-training on financial text?

From scratch

- Clean control of tokeniser, positional encoding, corpus weights
- Requires compute + data that *very few* organisations possess

Continue (DAPT)

- Reuses learned representations of a general foundation model
- Extends toward the domain at a fraction of the cost
- The dominant strategy in practice

UNDER THE HOOD

Definition. Given pre-trained parameters θ_0 of model M_0 and an unlabelled domain corpus $\mathcal{D}_{\text{domain}}$,

$$\theta^* = \arg \min_{\theta} \mathcal{L}_{\text{PT}}(\theta; \mathcal{D}_{\text{domain}}),$$

where $\mathcal{L}_{\text{PT}}$ is MLM (encoders) or CLM (decoders), the optimisation is *initialised* at θ_0 , and the learning rate is reduced ($\sim 10\times$) to mitigate catastrophic forgetting. Then fine-tune M^* on labelled task data.

- Gururangan et al. (2020): consistent downstream gains across 4 domains; *largest* when the domain is most dissimilar from pre-training.
- Finance is strongly out-of-distribution for web-text models $\Rightarrow$ DAPT gains should be large.
- Budget: typically **1–10B** domain tokens vs. hundreds of billions for original pre-training.

DAPT: How Much Data, and Two Refinements

Diminishing returns. The largest gain is in the first $\sim 1\text{B}$ domain tokens; smaller increments thereafter.

- Encoders (110–340M): **1–5B** domain tokens usually suffice.
- Decoders (7–70B): need proportionally larger corpora.

TAPT (task-adaptive)

Further pre-train on *unlabelled* examples from the task distribution before supervised fine-tuning. Consistent gains, but needs task-specific unlabelled data.

Catastrophic forgetting

Narrow-domain continuation can degrade general ability. Mitigate with a reduced LR, a large enough corpus, and **replay** — mixing a small fraction of general data back in.

Hirano (2024): Japanese financial LLMs confirm DAPT gains generalise across languages and sub-domains.

Corpus Composition: Sources for Financial Pre-training

Source	Scale	Quality	Access	Coverage
Bloomberg News	~300B	Very high	Proprietary	News, markets
SEC EDGAR	~50B	High	Open	Regulatory, corporate
Reuters TRC2	~2B	High	Licensed	Financial news
Financial PhraseBank	~5M	High	Open	Sentiment-labelled
Seeking Alpha	~10B	Medium	Open/API	Retail analysis
Earnings transcripts	~5B	High	Commercial	Corporate comms
SSRN / arXiv finance	~2B	High	Open	Academic research
General web (Pile)	~800B	Mixed	Open	Broad coverage

The defining fact

General web text (~800B) dwarfs even the largest *financial* source. This asymmetry is exactly *why DAPT beats from-scratch*: cheaply shift a model that already has the breadth, rather than rebuild it.

Example: A Central-Bank Communication Model

Goal: classify FOMC paragraphs as hawkish vs. dovish — needs the precise vocabulary of monetary policy (“accommodation”, “balance-sheet normalisation”, “neutral rate”).

A deliberately narrow corpus

- 1 FOMC statements + minutes 1994–present (~50M tokens) — target distribution.
- 2 Other central-bank publications: ECB, BoE, Beige Books (~200M).
- 3 Monetary-policy academic papers, SSRN/NBER (~500M).
- 4 Financial news mentioning policy keywords (1–2B).

This model would lose on general financial tasks — but on monetary-policy communication it *beats* a general model fine-tuned on the same labels, because pre-training aligns with both the vocabulary and the pragmatics of central-bank language.

The Standard Financial NLP Benchmarks

Bench	What it tests	Headline numbers
FinQA	8,281 QA pairs over 10-K/Q tables + prose; multi-step arithmetic. The hardest standard benchmark.	Fine-tuned specialists ~68–72% exec. acc.; zero-shot GPT-4 class ~55–60%.
FLUE	5-task NLU suite (GLUE analogue): sentiment, headline class., NER, FiQA QA, STS.	FinBERT > general BERT/RoBERTa; biggest margins on sentiment + NER.
FinBen	36 datasets, 24 tasks, 5 capability categories incl. generation + risk mgmt.	Designed to test <i>decoder</i> open-ended generation, not just classification.

How to read the headline numbers

Each benchmark stresses a different capability. FinQA isolates table + arithmetic reasoning; FLUE isolates lexical NLU; FinBen pushes open-ended generation. A model strong on one is *not* automatically strong on the others.

UNDER THE HOOD

Execution accuracy (FinQA):

$$\text{EA} = \frac{1}{|\mathcal{Q}|} \sum_{q \in \mathcal{Q}} \mathbf{1}[|\hat{a}_q - a_q| \leq \epsilon],$$

$\epsilon = 0$ for exact match, or a small relative threshold ($\sim 1\%$) for rounding.

Why a tolerance at all?

A model can execute the right reasoning program yet differ from the gold answer by a rounding step (e.g. reporting a ratio to two decimals). A small relative ϵ credits correct *reasoning* without rewarding spurious over-precision.

When Do FinLLMs Win? When Do They Lose?

FinLLMs win

Tasks tied to domain *lexical semantics*:

- PhraseBank sentiment — FinBERT $\gg$ BERT/RoBERTa, even after equal fine-tuning.
- EDGAR NER — SEC-BERT $\gg$ general BERT (entity surface forms differ).

General models win / tie

Tasks needing broad knowledge or deep reasoning:

- Macro / regulatory-history / geopolitical questions.
- FinQA with chain-of-thought on a large model often matches a smaller specialist — reasoning is a function of *scale*.

Domain advantage vs. model scale

The domain advantage is roughly *inversely proportional to model scale* at fixed difficulty.

- $\leq 7B$: DAPT gives large, consistent gains.
- $\geq 70B$: gains persist only for lexically sensitive tasks (sentiment, tone, NER); they fade for long-document reasoning.

A countervailing result

Rahimikia & Drinkall: year-specific FinLLMs can still beat far larger general models on return prediction — once one controls for look-ahead bias.

Two Caveats Every Benchmark Number Carries

Data contamination

A benchmark may already sit in a model's pre-training corpus. Several large models are *suspected* to have seen PhraseBank / FinQA train splits, inflating apparent performance. Rigorous evaluation needs held-out test sets with **temporal cutoffs** after the model's knowledge cutoff — which most published benchmarks do not enforce.

Hallucination is not solved by domain pre-training

It can be *worsened*: a model trained on financial news learns to generate plausible-but-unverified statistics. Kang & Liu (2023) find finance-trained models *more* likely to fabricate specific figures. Retrieval augmentation, constrained generation, and factual verification matter just as much for FinLLMs as for general models.

Deployment is Orthogonal to Benchmark Accuracy

THE BIGGER PICTURE

A model that tops FinQA but cannot be *licensed* commercially, needs a compute budget you cannot sustain, or produces outputs you cannot *trace* to source for audit — is not a deployable model.

Three consequential dimensions:

- 1 Licensing & data governance
- 2 Hosting strategy (cloud API vs. on-prem)
- 3 Cost-vs-performance $\Rightarrow$ hybrid architectures

Licensing, Hosting, Governance

Model	License
BloombergGPT	Proprietary, not released (terminal subscribers only)
FinBERT (Araci); FinBERT (Yang)	Apache 2.0 — commercial use OK
FinGPT	MIT
InvestLM / FinMA	Research-only (LLaMA-1 / LLaMA-2 license structures)

Data governance

GDPR / GLBA / CCPA constrain training on customer data. Customer balances, transactions, credit scores generally *cannot* train a model for *other* customers without consent. **Synthetic data** is a common workaround — at the cost of distribution shift.

Hosting. FinBERT (110–340M): single A10G, sub-50ms, fits in memory. 7–50B decoders: multi-GPU; **quantisation** (GPTQ, AWQ, QLoRA) cuts memory $2\text{--}4\times$ to fit 13–30B on one server. 50B+ : cluster-scale, few institutions.

Cost vs. Performance: the Effective-Cost Model

UNDER THE HOOD

Effective cost per correct output. With per-query model cost c_m , human review cost c_h , model correctness p_m , and routed fraction ρ :

$$C_{\text{eff}}(m, \rho) = \frac{c_m + \rho c_h}{p_m + \rho(1 - p_m)},$$

assuming human review recovers all errors. Minimise subject to $p_m + \rho(1 - p_m) \geq p^*$.

Hybrid routing pattern

A small, fast FinLLM handles the easy majority; low-confidence queries route to a larger model or a human. Effective when difficulty is **skewed** — a large “easy” head + a small “hard” tail.

Grounding instead of growing

A compact 7–13B decoder grounded in a well-indexed EDGAR store can match a larger ungrounded model on document tasks — at lower cost and with **full traceability** (an SR 11-7 / regulatory requirement).

- The retrieved passage pins generated figures to source text, cutting numerical hallucination.
- Traceability is not a nicety: model-risk regimes *require* an auditable link from output back to evidence.
- This makes a small grounded model a regulatory-grade alternative to a large ungrounded one.

Example: Cost-Optimal Regulatory-Filing Classifier

Task: classify “risk factor” sections of 10-Ks into 15 categories for 2,000 companies/year; 3–10 paragraphs each ($\approx 20,000$ – $60,000$ paragraphs).

Option	Accuracy	Annual cost
70B general model via API (all paragraphs)	93%	\$200–\$600
Fine-tuned FinBERT-base on a T4 GPU	88%	~\$400 server
Hybrid (FinBERT + route 12% low-conf. to 70B)	92%+	~\$250

Lesson

The hybrid dominates either model *alone* on the accuracy–cost frontier. The break-even point is highly sensitive to the *confidence distribution*, which must be measured on a held-out validation set — not assumed.

- ❶ **Register.** Financial language = specialised vocabulary + numeric density + tabular structure $\Rightarrow$ a systematic general-model gap. The LM dictionary quantifies the inversion starkly.
- ❷ **Two families.** Encoders (FinBERT, SEC-BERT) for fixed-label classification; decoders (BloombergGPT, FinGPT, FinMA, InvestLM) for open-ended generation and reasoning.
- ❸ **DAPT** is the broadly applicable strategy: continue the PT objective on a financial corpus blended with enough general text.
- ❹ **Benchmarks.** FinLLMs win on lexically sensitive tasks; lose the edge at large scale on reasoning. Watch contamination and hallucination.
- ❺ **Deployment.** License audit, governance, hosting, and the accuracy–cost trade-off. **Hybrid routing** dominates single-model deployments.

APPENDIX

Extended Material: Architectures, Sources, Mitigations

Extra / optional material — beyond the core lecture.

Appendix A — Encoder–Decoder Internals

Full encoder–decoder (T5) processing of a financial document:

- ➊ **Bidirectional encoder** attends over the entire 10-K source $\Rightarrow$ each source token sees full context.
- ➋ **Cross-attention** lets each decoder step query the encoded source.
- ➌ **Autoregressive decoder** emits the summary token by token.

Why factual consistency improves

Because the encoder consumes the *whole* source before generation starts, the decoder can copy figures and entities against a complete representation — fewer fabricated numbers in extraction-heavy summarisation than a pure decoder reading left-to-right.

Appendix B — Replay & Forgetting Detail

Catastrophic forgetting: continued pre-training on a narrow corpus degrades general language ability.

Three mitigations, increasing cost:

- ➊ **Reduced learning rate** ($\sim 10\times$ below original PT): cheapest; limits how far weights drift.
- ➋ **Sufficiently large domain corpus** relative to total tokens processed: keeps the update well-conditioned.
- ➌ **Replay**: interleave a small fraction of original general-corpus data with the domain corpus — modest extra cost, materially less forgetting.

TAPT can be stacked on top: DAPT (broad domain) $\rightarrow$ TAPT (task-distribution, unlabelled) $\rightarrow$ supervised fine-tuning.

Appendix C — Secondary Corpus Notes

- **Bloomberg archive:** decades of professionally edited news — consistent terminology, calibrated tone, high factual density. The gold standard, but proprietary (barrier for academia / open source).
- **SEC EDGAR:** free since 1996; full US public-company universe; continuously updated. Caveat: uneven quality (small-firm boilerplate) and XML structure needs careful parsing to split prose from tables/metadata.
- **News aggregators** (Reuters, AP, Seeking Alpha, subreddits): broader topical coverage, lower editorial quality. Reuters TRC2 is a standard source.
- **Analyst reports:** among the richest, most precise texts — almost entirely proprietary; licensing at pre-training scale is cost-prohibitive.

Appendix D — Quantisation for On-Prem Decoders

To fit a 13–30B decoder on a single high-memory GPU server:

Method	Idea	Effect
GPTQ	Post-training weight quantisation (4-bit)	2–4× memory reduction
AWQ	Activation-aware weight quantisation	Preserves salient channels
QLoRA	4-bit base + LoRA adapters in fine-tuning	Train + serve cheaply

All trade a modest accuracy cost for a large memory saving — the enabling technology for on-prem deployment of mid-scale decoders where data residency forbids a cloud API.

Appendix E — Hallucination Mitigation Stack

Domain pre-training does *not* remove hallucination. Production stack:

- ❶ **RAG**: ground generation in retrieved, verified passages.
- ❷ **Constrained / structured generation**: schema-enforced outputs.
- ❸ **Self-consistency**: K samples, majority vote (numerical reasoning).
- ❹ **Claim decomposition** + automated factual verification against source.
- ❺ **Confidence elicitation + abstention**: route low-confidence to humans.

Governance hook

SR 11-7 model-risk management and (forthcoming lecture) the EU AI Act / MiFID II make *traceability* a requirement — which is itself an argument for RAG-grounded architectures over ungrounded large models.